

mielke-geometrodynamics: residuals

equations measured 2026-09-05T13:17:30Z against the model of 2026-09-01

Corrected (3)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E08378 (was)	128	■ 0.136	$D\,D\,\vartheta^{\alpha}{}_{\beta}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}\wedge\vartheta^{\alpha}{}_{\beta}=0,\quad D\,\vartheta^{\alpha}{}_{\beta}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}\wedge\vartheta^{\alpha}{}_{\beta}=0,$	$DD\vartheta^{\alpha}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}=0,\quad D\,D\,\Pi_{\alpha}=-R_{\alpha}{}^{\beta}{}_{\gamma}{}^{\delta}\Pi_{\beta}\wedge\vartheta^{\gamma}{}_{\delta}=\pm\frac{i}{2}\ell^2\left(e_{\alpha}{}^{\beta}\Pi^{\beta}\right)\wedge\Sigma_{\beta}\cong0.$	$DD\vartheta^{\alpha}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}=0,\quad D\,D\,\Pi_{\alpha}=-R_{\alpha}{}^{\beta}{}_{\gamma}{}^{\delta}\Pi_{\beta}\wedge\vartheta^{\gamma}{}_{\delta}=\pm\frac{i}{2}\ell^2\left(e_{\alpha}{}^{\beta}\Pi^{\beta}\right)\wedge\Sigma_{\beta}\cong0.$
mielke-geometrodynamics_E08378 (now)	128	—	preserve the term Pontryagin index for the classification of real associated vector bundles (\textsc{Mayer} \& \textsc{Drechsler} 1977). Nevertheless, the denotation of \textsc{Belavin}	<i>preserve the term Pontryagin index for the classification of real associated vector bundles (Mayer & Drechsler 1977). Nevertheless, the denotation of Belavin</i>	$DD\vartheta^{\alpha}=R_{\beta}{}^{\alpha}{}_{\gamma}{}^{\delta}\vartheta^{\gamma}{}_{\delta}=0,\quad D\,D\,\Pi_{\alpha}=-R_{\alpha}{}^{\beta}{}_{\gamma}{}^{\delta}\Pi_{\beta}\wedge\vartheta^{\gamma}{}_{\delta}=\pm\frac{i}{2}\ell^2\left(e_{\alpha}{}^{\beta}\Pi^{\beta}\right)\wedge\Sigma_{\beta}\cong0.$
basis: inferred / ink					
mielke-geometrodynamics_E08482 (was)	134	■ 0.042	$\begin{array}{ll} q: & \underline{\Psi_{\parallel}(\vartheta)} = \underline{\Psi_{\parallel}(\vartheta)}, \\ \pm p: & \underline{\Pi_{\alpha}}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta), \end{array}$	$q: \quad \underline{\Psi_{\parallel}(\vartheta)} = \underline{\Psi_{\parallel}(\vartheta)}, \\ \pm p: \quad \underline{\Pi_{\alpha}}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta),$	$q: \quad \underline{\Psi_{\parallel}(\vartheta)} = \underline{\Psi_{\parallel}(\vartheta)}, \\ \pm p: \quad \underline{\Pi_{\alpha}}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta),$
mielke-geometrodynamics_E08482 (now)	134	—	$\begin{array}{ll} q: & \underline{\Psi_{\parallel}(\vartheta)} = \underline{\Psi_{\parallel}(\vartheta)}, \\ \pm p: & \underline{\Pi_{\alpha}}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta), \end{array}$	$q: \quad \underline{\Psi_{\parallel}(\vartheta)} = \underline{\Psi_{\parallel}(\vartheta)}, \\ \pm p: \quad \underline{\Pi_{\alpha}}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta),$	$q: \quad \underline{\Psi_{\parallel}(\vartheta)} = \underline{\Psi_{\parallel}(\vartheta)}, \\ \pm p: \quad \underline{\Pi_{\alpha}}(\vartheta) = -i\ell^2 \frac{\delta}{\delta \underline{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta),$
basis: inferred / ink					
mielke-geometrodynamics_E08972 (was)	314	■ 0.153	$\alpha:=\alpha\wedge\alpha\wedge\alpha$	$D\alpha:=d\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega$	$D\alpha:=d\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega$
mielke-geometrodynamics_E08972 (now)	314	—	$\alpha:=\alpha\wedge\alpha\wedge\alpha$	$D\alpha:=d\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega$	$D\alpha:=d\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega$
basis: inferred / ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_F00431	100	■ 0.590	$\mathrm{t}_{\xi}:=\xi \mathrel{\mathop{\rightarrow}\limits^{\mathsf{D}}}\xi$	(not rendered)	covariant Lie derivative $\mathbb{L}_{\xi}:=\xi\lrcorner D+D\xi\lrcorner$ on M with respect to an arbitrary vector
mielke-geometrodynamics_E00472 (7.3.11)	155	■ 0.682 ● C +23	$\stackrel{+}{\Omega}^{\{t\}}=0\Longleftrightarrow R_{ij}^{\{\backslash\backslash\}}=0\Longleftrightarrow R_{ij}^{\{\backslash\backslash\}}=\Lambda^{\prime\prime}\mathfrak{g}_{ij}.$	(not rendered)	$\overset{+}{\Omega}^{\{t\}}=0\Longleftrightarrow \mathcal{R}_{ij}^{\{t\}}=0\Longleftrightarrow R_{ij}^{\{t\}}=\Lambda'g_{ij}.$
mielke-geometrodynamics_E00857 (12.4.4)	278	■ 0.925 ● C +3	$\left.\mathbf{t}_{\mathfrak{n}}\right.\mathscr{G}^A\cong n\right\lfloor D\underline{\tau}^A\right\rfloor,$	(not rendered)	$\mathbb{L}_n\mathscr{G}^A\cong n\lrcorner(D\underline{\tau}^A),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Low confidence

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_101 EQ0293 (4.5.4) [conf 0.052]	101	■ 0.052 ● N +0	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^\alpha \right) \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi}.$	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^\alpha \right) \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi}.$	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^\alpha \right) \wedge \frac{\partial L}{\partial \vartheta^\alpha} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi}.$
mielke-geometrodynamics_47 EQ0064 (2.8.6) [conf 0.054]	47	■ 0.054 ● N -2	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr}(\Omega) \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$
mielke-geometrodynamics_133 EQ0400 (6.4.19) [conf 0.060]	133	■ 0.060 ● S +1	$\stackrel{(\pm)}{\mathcal{G}}_\alpha := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_\alpha = \stackrel{(\pm)}{D} \stackrel{(\pm)}{\underline{A}}_\alpha = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_\alpha := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_\alpha = \stackrel{(\pm)}{D} \stackrel{(\pm)}{\underline{A}}_\alpha = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_\alpha := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_\alpha = \stackrel{(\pm)}{D} \stackrel{(\pm)}{\underline{A}}_\alpha = 0,$
mielke-geometrodynamics_229 EQ0718 (10.9.4) [conf 0.066]	229	■ 0.066 ● N +0	$\begin{aligned} & \Sigma_\alpha := e_\alpha \downharpoonleft L_\Psi - (e_\alpha \downharpoonleft \Psi) \wedge \frac{\partial L_\Psi}{\partial \Psi} - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi} \\ & \quad - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi} - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi}; \end{aligned}$	$\Sigma_\alpha := e_\alpha \downharpoonleft L_\Psi - (e_\alpha \downharpoonleft \Psi) \wedge \frac{\partial L_\Psi}{\partial \Psi} - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi} - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi};$	$\begin{aligned} & \Sigma_\alpha := e_\alpha \downharpoonleft L_\Psi - (e_\alpha \downharpoonleft \Psi) \wedge \frac{\partial L_\Psi}{\partial \Psi} - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi} \\ & \quad - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi} - (e_\alpha \downharpoonleft D\Psi) \wedge \frac{\partial L_\Psi}{\partial D\Psi}; \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_131 E08391 (6.4.6) [conf 0.066]	131	0.066	$\begin{aligned} \Sigma_{\alpha} &:= \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ &\quad - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}. \end{aligned}$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - \left(e_{\alpha} \rfloor R_{\beta}^{\gamma} \right) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$	$\Sigma_{\alpha} := \frac{\delta L}{\delta \vartheta^{\alpha}} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D\Psi) \wedge \frac{\partial L}{\partial D\Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ - (e_{\alpha} \rfloor T^{\beta}) \wedge \frac{\partial L}{\partial T^{\beta}} + D \frac{\partial L}{\partial T^{\alpha}} - (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \frac{\partial L}{\partial R_{\beta}^{\gamma}}.$
mielke-geometrodynamics_168 E08529 (7.7.49) [conf 0.099]	168	0.099	$\begin{aligned} L_{\text{Euler}} &:= \frac{(-1)^{\text{sig}+1}}{2} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ &= dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta^{(*)}} - \frac{1}{3} \Gamma_{\alpha}^{\beta^{(*)}} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ &\equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \\ &\equiv -L_{\text{SKY}} \\ &\quad - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \right) \end{aligned}$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \left\{ \Omega \wedge \Omega^{(*)} \right\} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta^{(*)}} - \frac{1}{3} \Gamma_{\alpha}^{\beta^{(*)}} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \\ \equiv -L_{\text{SKY}} \\ - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} - \frac{1}{4} \text{Ric}_{\alpha}^{\alpha} \wedge {}^*\text{Ric}_{\beta}^{\beta} \right)$	$L_{\text{Euler}} := (-1)^{\text{sig}+1} \text{Tr} \{ \Omega \wedge \Omega^{(*)} \} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta^{(*)}} - \frac{1}{3} \Gamma_{\alpha}^{\beta^{(*)}} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} + \frac{1}{2} \text{Ric}_{\alpha}^{\beta^{(*)}} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \\ \equiv -L_{\text{SKY}} \\ - 2 \left(\text{Ric}_{\alpha\beta} \wedge {}^*\text{Ric}^{\alpha\beta} - \frac{1}{3} \Gamma_{\alpha}^{\beta^{(*)}} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right)$
mielke-geometrodynamics_315 E08974 (14.2.15) [conf 0.099]	315	0.099 N +0	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\circ\Theta} = 0 \right],$	$\Theta^{\perp} := \pi(\hat{\Theta} - [\hat{\alpha}, \hat{\vartheta}]) = \left[\frac{\Theta}{0} \middle \frac{\hat{\ell}\Omega'}{\pi\overset{\circ}{\Theta} = 0} \right],$
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					